

HEAVEN'S LIGHT IS OUR GUIDE



RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF MECHATRONICS ENGINEERING, 21Series

Course No: MTE 3101

Course Title: Control Systems Sessional

Experiment No: 04

Experiment Name: Determination of controllability and observability of a system using both transfer function and state space.

Submitted To:

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Experiment No: 04

Experiment Name: Determination of controllability and observability of a system using both transfer function and state space.

Objectives:

1. Understand the concepts of controllability and observability in control systems.
2. Analyze a system's controllability and observability using: Transfer function & State-space representation

Theory:

Controllability and observability are key principles in control system analysis and design. Inputs and outputs are vital in defining a system's ability to be fully controlled and accurately infer internal conditions. This report focuses on: Two methodologies are used to determine controllability and observability in dynamic systems: the transfer function method and the state-space representation.

Controllability: If an input to a system can be found that takes every state variable from a desired initial state to a desired final state, the system is said to be controllable; otherwise, the system is uncontrollable. In order to be able to determine controllability or, alternatively, to design state feedback for a plant under any representation or choice of state variables, a matrix can be derived that must have a particular property if all state variables are to be controlled by the plant input, u . We now state the requirement for controllability, including the form, property, and name of this matrix. An n th-order plant whose state equation is

$$\dot{x} = Ax + Bu$$

is completely controllable if the matrix

$$C_M = [B \ AB \ A^2B \ \dots \ A^{n-1}B]$$

is of rank n , where C_M is called the controllability matrix.[1]

Observability: Observability is the ability to deduce the state variables from a knowledge of the input, $u(t)$, and the output, $y(t)$. Pole placement for an observer is a viable design technique only for systems that are observable.

In order to determine observability for systems under any representation or choice of state variables, a matrix can be derived that must have a particular property if all state variables are to be observed at the output. We now state the requirements for observability, including the form, property, and name of this matrix. An n th-order plant whose state and output equations are, respectively,

$$\dot{x} = Ax + Bu$$

$$y = Cx$$

is completely observable if the matrix

$$O_M = \begin{bmatrix} C \\ CA \\ \dots \\ CA^{n-1} \end{bmatrix}$$

is of rank n , where O_M is called the observability matrix.[1]

Required Apparatus:

MATLAB

MATLAB Code:

1. Using State Space

```
clc;
clear all;
clf;

% Define State-Space Matrices
A = [10 15; -1 0];
B = [1; 2];
C = [0 10];
D = 0;

% Create State-Space System
sys1 = ss(A, B, C, D);

%Check Controllability of system1
if rank(ctrb(A,B))==rank(A)
disp("System 1 is Contrrollable");
else
disp("System 1 is not Contrrollable");
end;

%Check Obsarvability
if rank(observ(A,C))==rank(A)
disp("System 1 is Obsarvable");
else
disp("System 1 is not Obsarvable");
end;

% Define State-Space Matrices
A1 = [10 15 0; 0 1 0; -8 -10 9];
B1 = [0; 0; 1];
C1 = [10 -8 0];
D1 = 0;

% Create State-Space System of
system2
sys2 = ss(A1, B1, C1, D1);

%Check Controllability
if rank(ctrb(A1,B1))==rank(A1)
disp("System 2 is Contrrollable");
else
disp("System 2 is not Contrrollable");
end;

%Check Obsarvability
if rank(observ(A1,C1))==rank(A1)
disp("System 2 is Obsarvable");
else
disp("System 2 is not Obsarvable");
end;
```

2. Using Transfer Function

```
clc;
clear all;
clf;

%system Parameters
J=0.01;
b=0.1;
K=0.01;
R=1;
L=0.5;
s= tf('s');
```

```
%Transfer function Representation
TF=K/((J*s+b)*(L*s+R)+K^2);
```

```
%Convert TF to State Space
[num, den] = tfdata(TF, 'v');
[A,B,C,D] = tf2ss(num,den);
```

```
%Check Controllability
if rank(ctrb(A,B))==rank(A)
disp("System is Contrallable");
else
disp("System is not Contrallable");
end;
```

```
%Check Obsarvability
if rank(observ(A,C))==rank(A)
disp("System is Obsarvable");
else
disp("System is not Obsarvable");
end;
```

```
%system parameters
```

```
M= 0.012;
D= 0.19;
K1= 0.5;
K2= 0.5;
s=tf('s');
```

```
%Transfer Function Repestantion
```

```
TF= 1/(M*s^2+D*s+K1+K2);
%Convert TF to State Space
[num, den] = tfdata(TF, 'v');
[A1,B1,C1,D1] = tf2ss(num,den);
```

```
%Check Controllability
if rank(ctrb(A1,B1))==rank(A1)
disp("System2 is Contrallable");
else
disp("System2 is not Contrallable");
end;
```

```
%Check Obsarvability
if rank(observ(A1,C1))==rank(A1)
disp("System2 is Obsarvable");
else
disp("System2 is not Obsarvable");
end;
```

Result:

```
Command Window
New to MATLAB? See resources for Getting Started.

System 1 is Contrallable
System 1 is Obsarvable
System 2 is not Contrallable
System 2 is not Obsarvable
fx >> |
```

(a)

```
Command Window
New to MATLAB? See resources for Getting Started.

System is Contrallable
System is Obsarvable
System2 is Contrallable
System2 is Obsarvable
fx >> |
```

(b)

Figure 1: Controllability and observability test using (a) state space (b) transfer function

Discussion:

This experiment focused on analyzing a system's controllability and observability using both transfer function and state-space approaches. The state-space method provided a matrix-based test using controllability and observability matrices. If these matrices had full rank, the system was controllable and observable. The transfer function approach helped identify whether the system was minimal—i.e., free from pole-zero cancellations that might affect control or observation. The comparison highlighted that while the transfer function is useful for a quick check, state-space analysis gives deeper insight into system dynamics.

Conclusion:

The experiment successfully demonstrated how to verify the controllability and observability of a system using both theoretical and practical methods. The state-space method proved more comprehensive, especially for systems with multiple states. Understanding these concepts is crucial for designing reliable control systems.

Reference:

[1] N. S. Nise, *Control Systems Engineering*, 6th ed. Hoboken, NJ, USA: John Wiley & Sons, 2011.